

BANJA LUKA, Bosnia and Herzegovina

WMO: 145420

Lat: **44.7938N**Lon: **17.2058E**Elev: **512**StdP: **14.43**Time Zone: **1.00 (EUC)**Period: **02-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	13.5	18.3	10.1	9.4	15.3	14.7	11.8	21.3	19.0	55.1	15.0	50.1	2.1	290	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	96.3	71.4	92.7	71.1	89.4	70.3	74.4	89.7	72.7	88.3	71.2	86.5	5.0	20

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.5	110.5	82.2	67.7	103.8	80.1	66.2	98.3	77.7	38.2	89.9	36.6	88.5	35.3	86.6	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 15.7	11.9	9.1	DB	4.6	100.8	8.0	4.1	-1.1	103.8	-5.8	106.1	-10.3	108.4	-16.1	111.4	(4)
(5)			WB	4.3	76.8	7.7	1.5	-1.2	77.9	-5.7	78.8	-10.0	79.6	-15.6	80.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	54.7	35.2	37.1	45.6	55.6	62.5	70.2	73.7	73.1	63.8	54.7	46.8	37.1
	DBStd	15.61	9.16	9.35	9.06	6.22	6.85	6.96	5.98	6.17	7.06	7.31	8.43	8.11
	HDD50	1649	465	367	190	23	3	0	0	0	0	37	155	409
	HDD65	4612	924	781	601	287	130	33	6	9	105	325	546	865
	CDD50	3362	5	6	54	189	389	605	733	715	414	183	60	9
	CDD65	851	0	0	0	4	51	187	274	259	69	6	1	0
	CDH74	9735	0	0	9	130	659	2071	3171	2837	757	97	4	0
	CDH80	4373	0	0	0	22	176	881	1568	1438	280	8	0	0
Wind	WSAvg	3.2	2.9	3.2	4.1	3.7	3.5	3.2	3.3	3.4	2.8	2.8	2.9	2.9
Precipitation	PrecAvg	41.6	2.8	2.7	2.7	3.6	3.9	4.3	4.1	3.8	3.4	2.8	3.9	3.5
	PrecMax	50.4	5.3	7.7	6.0	5.8	7.6	10.9	8.8	12.0	8.1	10.9	7.7	7.5
	PrecMin	27.0	0.4	0.6	0.6	1.0	0.6	2.3	1.4	0.2	0.7	0.0	1.0	0.3
	PrecStd	5.7	1.3	1.6	1.4	1.2	1.9	2.2	1.9	2.5	2.0	2.2	1.7	1.7
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.0	66.7	75.2	83.4	88.7	95.3	100.6	102.1	94.2	81.2	73.5	65.8
		MCWB	51.5	52.1	55.4	61.3	67.9	73.3	72.5	71.0	68.7	63.5	58.5	54.9
	2%	DB	58.8	60.9	70.9	77.5	84.8	91.9	96.3	96.9	87.1	77.1	69.7	61.3
		MCWB	49.2	49.4	53.2	58.5	67.0	71.9	71.7	70.8	66.7	62.2	56.9	51.3
	5%	DB	54.2	56.8	66.8	73.7	81.1	88.7	92.6	92.6	82.1	73.1	64.8	56.0
		MCWB	46.6	47.3	51.9	57.2	65.0	70.7	71.6	70.2	65.5	60.9	54.5	48.2
	10%	DB	48.5	52.0	62.3	69.7	77.4	85.3	88.9	88.5	77.7	68.5	60.1	48.9
		MCWB	43.4	45.2	50.1	55.5	63.2	69.2	70.8	69.3	64.4	59.1	53.0	44.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.0	52.9	56.7	63.2	70.9	76.2	76.8	75.9	70.8	65.8	60.0	55.6
		MCDB	62.2	65.4	71.3	79.8	85.6	91.0	91.3	90.9	86.1	77.3	69.7	64.8
	2%	WB	49.6	50.1	54.5	60.1	68.6	73.7	74.7	73.4	68.6	63.4	57.8	51.7
		MCDB	57.8	59.3	67.7	73.5	82.1	88.0	89.9	89.9	82.9	73.7	66.5	60.3
	5%	WB	47.1	47.9	52.6	58.2	66.2	71.8	73.0	71.7	66.9	61.5	56.0	48.3
		MCDB	53.6	55.5	64.4	69.9	78.1	86.3	88.6	88.7	79.2	71.0	64.1	54.8
	10%	WB	43.8	45.6	50.8	56.6	64.0	70.1	71.4	70.2	65.2	59.7	53.3	45.0
		MCDB	48.2	51.5	61.3	67.2	75.2	83.1	86.7	85.5	75.1	67.8	59.8	48.8
Mean Daily Temperature Range	5% DB	MDBR	13.8	14.8	19.8	21.9	21.5	21.6	23.4	23.6	20.6	19.5	16.2	12.8
		MCDBR	23.3	25.9	30.8	30.9	28.6	28.3	30.8	32.9	29.6	27.3	25.7	23.7
		MCWBR	14.8	15.5	15.9	14.7	13.1	11.6	11.0	11.5	12.3	14.2	14.8	14.8
	5% WB	MCDBR	22.0	23.6	26.9	27.2	26.2	26.3	27.2	28.5	25.4	25.3	22.6	21.8
	MCWBR	14.1	14.5	14.3	13.6	12.7	11.4	11.2	11.3	11.2	13.8	13.4	13.9	
Clear-Sky Solar Irradiance	taub	0.328	0.357	0.392	0.443	0.433	0.432	0.432	0.432	0.416	0.386	0.349	0.329	
	taud	2.398	2.311	2.254	2.132	2.212	2.257	2.254	2.260	2.279	2.360	2.416	2.414	
	Ebn at Noon	247	259	265	259	266	267	265	260	254	245	236	234	
	Edh at Noon	24	31	38	46	44	43	42	40	36	30	23	22	
All-Sky Solar Radiation	RadAvg	407	618	993	1346	1668	1896	1938	1725	1181	803	473	340	
	RadStd	57	90	138	167	149	159	138	193	146	96	57	61	

Historical Trends

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